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LETTER

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First-Principles Study of Lithium Borocarbide as a Cathode Material for Rechargeable Li ion Batteries

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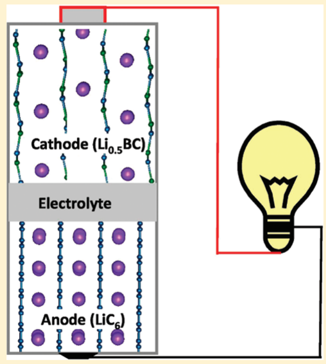
ABSTRACT: Computational simulations within density functional theory are performed to investigate the potential application of a lithium borocarbide (LiBC) compound as a unique material for lithium ion batteries. The graphene-like BC sheets are predicted to be Li^+ intercalation hosts with the Li ion capacity surprisingly surpassing that of graphite. Here, the layered Li_xBC structure is preserved with $x \geq 0.5$, indicating that half of the Li ions in the LiBC compound are rechargeable. Furthermore, the intercalation potential (equilibrium lithium-insertion voltage of 2.3−2.4 V relative to lithium metal) is significantly higher than that in graphite, allowing $Li_{0.5}BC$ to function as a cathode material. The reversible electrochemical reaction, $LiBC \rightleftharpoons Li_{0.5}BC + 0.5Li$, enables a specific energy density of 1088 W h/kg and a volumetric energy density of 2463 W h/L. The volume change is less than 3% during the charging and discharging process. This discovery could lead to the development of a unique high-capacity LiBC Li ion cathode material.



SECTION: Energy Conversion and Storage

Lithium ion (Li ion) batteries are believed to hold the promise of high-capacity energy storage for fully electric vehicles (EVs). However, to date, it is still necessary to develop advanced electrode materials for EVs as none of the known electrode materials simultaneously satisfy the crucial requirements including high-energy density, good electronic and ion conductivity to enable high-rate capability, tolerable volume expansion with simultaneous high Li loading, and durable phase transformation allowing long-cycle life and competitively low cost.[1,2] Although Li−sulfur batteries with very high energy density have been demonstrated recently,[3] the requirement of using mesoporous carbon material as a sulfur host reduces the volumetric capacity substantially.

Graphite is highly conductive, lightweight, stable, inexpensive, and environmentally benign and is currently the anode material of choice for battery-powered electronic devices as well as first-generation Li ion hybrid EVs. The very low open-circuit voltage (V_{OC}) of ~0.2 V relative to Li/Li^+ allows graphite to function only as an anode for high-power applications. Currently, the most common cathode materials are transition-metal oxides and phosphates. Although these materials have high V_{OC} between 3.0 and 4.5 V relative to Li/Li^+, they suffer from many disadvantages such as high cost, inferior electronic conductivity, and low energy densities (when coupled with graphite) of 400−700 W h/kg as well as low gravimetric capacities of only 140−180 mA h/g.[2,4−9]

Here, we explore the possibility of developing a new cathode material with a graphite-like structure that has both superior electronic conductivity and a significantly higher operating voltage relative to Li/Li^+. To increase the operating voltage, it is possible to employ substitutional boron atoms in the graphite lattice, creating substantial electron deficiency.[10,11] Graphitic BC_3 has been reported previously,[12−15] but experimental characterization of the crystalline BC_3 material is extremely limited. On the other hand, however, lithium borocarbide (LiBC, with the crystal structure shown in Figure 1) was first synthesized in 1995 by Wörle et al.,[16] and the Li_xBC compounds with various x values have been synthesized by many groups in bulk quantities.[17−23] Although it was claimed that 75% of the Li content may be extracted from the LiBC without altering the layered structure,[16] recent experimental evidence shows that the Li_xBC structures are sufficiently durable only at $x > 0.5$,[20,22] and phase transition happens at less Li content.[18]

Most of the above studies focus on the potential superconductivity of the Li_xBC compounds.[24] However, to our best knowledge, no one has studied the electrochemical properties of Li_xBC. Here, we employ density functional theory (DFT) to probe the reversible Li insertion/extraction process for LiBC and its relevance to the discovery of a novel Li intercalation compound. Because of the electron deficiency of B, the calculated V_{OC} is increased to 2.3−2.4 V, making the material viable for a high-capacity cathode.

Lithium borocarbide forms a hexagonal layered structure with a symmetry described by the space group of $P6_3/mmc$ (D_{6h}^4), as shown in Figure 1. Its atomic structure is similar to that of MgB_2, except that the lattice constants of the latter are ~10% larger.[25] The BC layers are stacked along the c-axis with the B and C atoms alternately distributed both within a BC layer and in the atomic columns along the c-axis. The lithium ions are intercalated in the BC layers and form atomic columns (along the c-axis) going through the centers of the hexagonal BC rings.

In the BC sheets, the B−C bonding remains in an sp^2 configuration with each π-bonding orbital occupied by a C $2p_z$ electron and a Li 2s electron transferred to the B $2p_z$ orbital. The band structure and density of states (DOS) of LiBC are shown in

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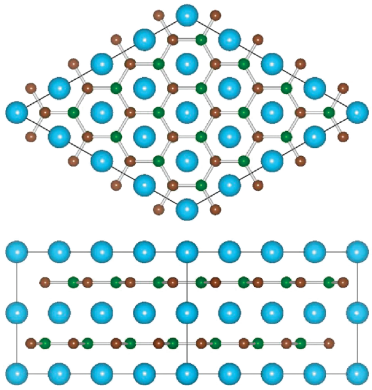

Figure 1. Top and side views of the layered crystal structure of LiBC in a $4 \times 4 \times 1$ super cell. The B (green) and C (dark brown) atoms form graphene-like sheets, which are intercalated with Li atoms (light blue). The interlayer distance is 3.415 Å.

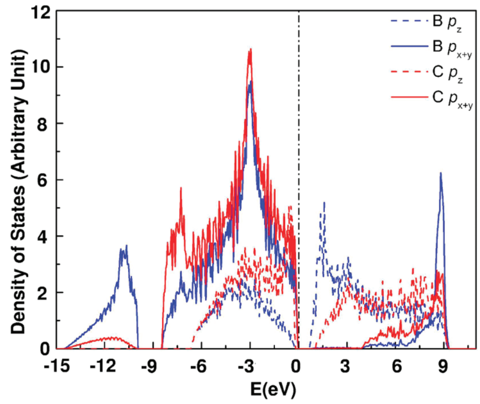

Figure 3. Calculated partial DOS for B and C atoms in the layered LiBC crystal structure. The dashed vertical line marks the position of the Fermi level.

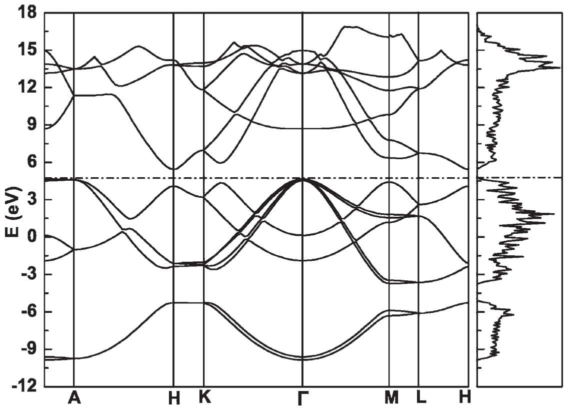

Figure 2. Calculated band structure and DOS for LiBC. The dash–dot line is the Fermi level.

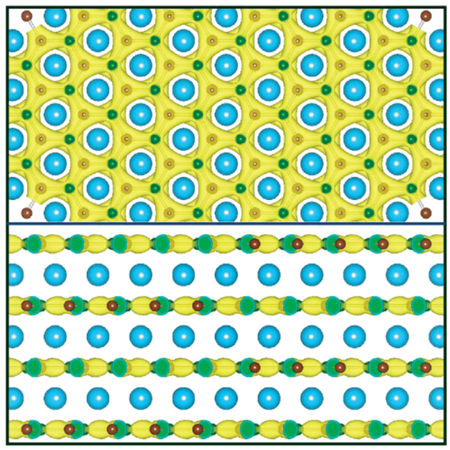

Figure 4. Top view (top) and side view (bottom) of the partial charge densities of LiBC at Γ point.

Figure 2. The valence band maximum (VBM) appears at the Γ point, and the conduction band minimum (CBM) is at the H point, which gives an indirect band gap of 0.78 eV. This calculated band gap is smaller than the experimental value (>1.0 eV)[26] because of the well-known local density approximation (LDA) underestimation of band gaps. To identify the nature of the bands, we calculated the partial density of state (PDOS) of LiBC. Figure 3 shows that the VBM edge of LiBC (at the Γ point in Figure 2) has the character of bonding σ orbitals. The CBM edge (at the H point in Figure 2) contains only the $2p_z$ state of boron atoms, that is, antibonding π orbitals. These features are also clearly shown in the partial charge density of VBM of LiBC at the Γ point (Figure 4). In addition, a polar effect is observed as the C p_z orbitals are occupied with more charge than the B p_z orbitals.

Now, we assess LiBC as a potential cathode material for rechargeable lithium batteries. Three key parameters should then be evaluated, (1) the maximum percentage of reversibly extractable Li from LiBC, m, with the layered structure of Li_xBC being stable over the entire range of $(1 - m) < x < 1$; (2) the voltage profile; and (3) the corresponding energy density. The insertion/extraction reaction that occurs in the cathode material of a rechargeable Li ion battery is expressed as

$$Li_xBC \rightleftharpoons Li_{x-\Delta x}BC + \Delta x Li^{bulk} \qquad (1)$$

where Li^{bulk} indicates bulk lithium metal. The average open-circuit voltage (V_{OC}) depends on the difference of the Li chemical potential in LiBC $\mu_{Li}^{(c)}$ and bulk lithium metal $\mu_{Li}^{(a)}$ as follows[27]

$$V_{OCC} = \frac{\mu_{Li}^{(c)} - \mu_{Li}^{(a)}}{zF} \qquad (2)$$

where F is the Faraday constant and z is the electronic charge of lithium ions in the electrolyte (for Li, $z = 1$). For the LiBC cathode, the voltage profile versus Li/Li^+ is determined by[27]

$$V_{OC}^{LiBC}(x) = \frac{E(Li_{x-\Delta x}BC) - E(Li_xBC) + \Delta x E(Li)}{e\Delta x} \qquad (3)$$

where $E(Li_xBC)$ and $E(Li)$ denote the total energy of Li_xBC and lithium metal, respectively. Practically, the V_{OC} should be slightly different from the theoretical value with the complex effect of electrode–electrolyte interfaces considered.[28]

In order to determine the maximum reversibly extractable Li percentage m, we test the stability of Li_xBC at x = 0.00, 0.25, 0.50, and 0.75 through structure optimization and molecular dynamics (MD) simulation. To test the stability, we use in the MD simulation a unit cell size 4 times larger than that used in structure optimization. In the structure optimization for all of these

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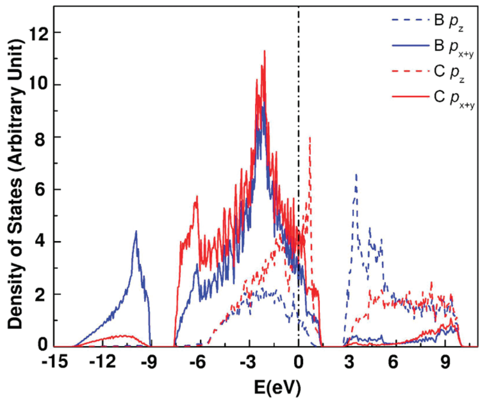

Figure 5. Calculated PDOS for B and C atoms in the $Li_{0.5}BC$ compound. The dashed vertical line marks the position of the Fermi level.

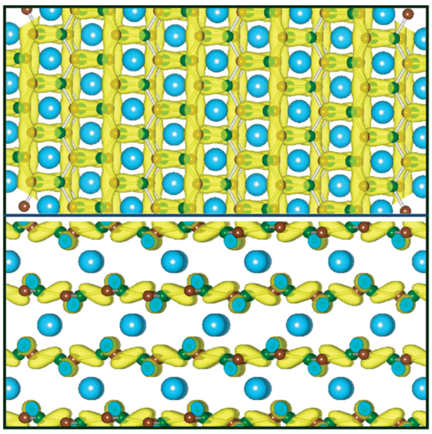

Figure 6. Top view (top) and side view (bottom) of partial charge densities of $Li_{0.5}BC$ near the Fermi level. Buckling of the BC sheets can be seen. The interlayer distance is 3.471 Å. The C, B, and Li atoms are colored in the same way as those in Figure 1.

unsaturated compounds, we found that uniform distribution of the lithium ions always gives the lowest energy due to the minimization of Coulomb repulsion. Subsequent MD simulations at 500 K for 4 ps (8000 MD steps) show that the layered structure of Li_xBC can be well preserved at $x \geq 0.25$. However, when $x < 0.25$, strong covalent interlayer B−C bonds are formed so that the Li^+ intercalation/extraction is irreversible. Although our MD simulation confirms the early experimental observation that $Li_{0.25}BC$ is stable, recent experimental evidence shows that $Li_{0.5}BC$ is much more durable against phase transition.[18,20,22] Therefore, we assume here that the maximum reversibly extractable Li quantity in LiBC is 0.5 Li per BC pair. Figures 5 and 6 show, respectively, the PDOS and partial charge density of $Li_{0.5}BC$. In contrast to the LiBC, the highest-occupied state of $Li_{0.5}BC$ at the Fermi level is comprised of roughly half π and half σ components (Figure 5). Such a rehybridization leads to BC sheet buckling (Figure 6) and a slight reduction of the B−C bond length from 1.59 Å in LiBC to 1.53−1.55 Å in $Li_{0.5}BC$. The rehybridization and buckling stabilize the $Li_{0.5}BC$ structure. All of these observations agree with previous theory.[29]

Table 1 shows the average open-circuit voltages (V_{OC}) of Li_xBC calculated from eq 3 with x = 0.25, 0.50, and 0.75, and Δx = 0.25. During lithium intercalation, the V_{oc} gradually decreases from 2.55 V at x = 0.25 to 2.42 V at x = 0.50 and further to 2.32 V at x = 0.75. Overall, this shows very good linearity of the V_{OC} behavior. The specific (volumetric) energy density corresponding to the reversible electrochemical reaction LiBC $\rightleftharpoons$ $Li_{0.5}BC$ + 0.5Li is calculated to be 1088 W h/kg (2463 W h/L). Therefore, LiBC is a promising cathode candidate for energy storage, with a high reversible capacity. Furthermore, the volume change caused by the charge/discharge of Li ions is less than 3%, as shown in Table 1, indicating that the material will be highly rechargeable.

Table 1. Calculated Volume Change and Open-Circuit Voltage of Li_xBC

x	0.25	0.50	0.75	1.00
$\Delta v/v$	−4.89%	−2.41%	−0.49%	0.00
V_{OC} (V)	2.55	2.42	2.32	

The last issue concerning the application of LiBC as an electrode material for Li ion batteries is the electronic and ionic conductivity. Although an ideal LiBC crystal has a moderate band gap, good electron conductivity can still be achieved because all of the Li_xBC ($x < 1$, i.e., with Li vacancies) compounds are good electron conductors. According to experiment, the activation energy of the onset of conductivity in LiBC is only 18 meV,[22] indicating that Li vacancies always emerge in normal fabrication processes. On the other hand, the ionic conductivity can be measured by the diffusion barrier of Li ions inside of Li_xBC. We found that the diffusion pathway encounters its transition point as the Li ion passes over the BC bond between the initial and final equilibrium ion positions. The Li ion diffusion barrier in $Li_{0.5}BC$ is calculated to be 0.4 eV. However, the barrier of Li ion diffusion from LiBC to $Li_{0.5}BC$ across the interface is 0.28 eV. These values are comparable with the barrier of 0.3−0.49 eV for Li ion diffusion in LiC_6.[30] The low diffusion barrier implies a high power density if the ion conductivity in electrolytes is high enough.

In conclusion, we have calculated the electronic structures of Li_xBC ($0 < x \leq 1$) using the first-principles methods based on density functional theory. The layered BC structures can be well preserved after half of the intercalated lithium atoms are extracted out of the LiBC host. The volume change of the cell is ∼3%. Exceptionally high specific energy density of 1088 W h/kg and a volumetric energy density of 2463 W h/L are predicted with an open circuit voltage of 2.3−2.4 V relative to Li/Li^+. Compared to the recently proposed magnesium boride electrode materials for Mg ion batteries,[31] both the operating voltage and energy density are improved. This work is also relative to significantly broader applications as the unsaturated lithium borocarbide compound Li_xBC may be used as a precursor to synthesize BC materials for metal dispersion, which are important for hydrogen storage and catalysis.[10,11]

■ COMPUTATION SECTION

All calculations are carried out within DFT methods, as implemented in the Vienna ab initio simulation package (VASP).[32] A plane-wave expansion up to 500 eV is employed in combination with an all-electron-like projector augmented wave (PAW) potential.[33] Exchange-correlation is treated within the LDA using the functional parametrized by Ceperley and Alder.[34] For lithium atoms, the chosen potential treats its two 1s and one 2s electrons as valence electrons. The computational unit cells are constructed in such a way that Li atoms intercalate between the hexagonal BC sheets parallel to the $x-y$ plane, and the LiBC layers stack periodically in the z direction. Each unit cell contains two BC layers. Periodic boundary conditions are applied to the unit cell in all three dimensions. The Brillouin zone integrations are performed using Monkhorst−Pack-type meshes,[35] with the mesh sizes selected according to the sizes of the unit cells. For the primitive cell of LiBC containing 2 Li/B/C atoms, the 11 × 11 × 11 mesh of k-points is chosen, whereas for a 4 × 4 × 1 super cell (Figure 1) containing 32 Li/B/C atoms, the 5 × 5 × 5 k-point mesh is found to be sufficient. The structures are considered to be in equilibrium when the maximum force converges below 0.01 eV/Å. The optimized lattice constants of LiBC, a = 2.718 Å and c = 6.831 Å, are in good agreement with experimental results of a = 2.752 Å and c = 7.058 Å.[13]

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